

Experiment No - 1

B.M.I.T

Name of Experiment

To Find the Coefficient of discharge for given Venturimeter

Date :-

Aim :-

To Find out Coefficient for given venturimeter

Apparatus :-

The unit consists of Venturimeter having inlet dia. d_1 & throat dia. d_2 in piping system. The pressure tapping at inlet are connected to different manometer. The unit is provided flow control valve with which flow through venturimeter can be varied. A collecting tank with glass gauge is provided with which actual discharge can be measured.

Theory :-

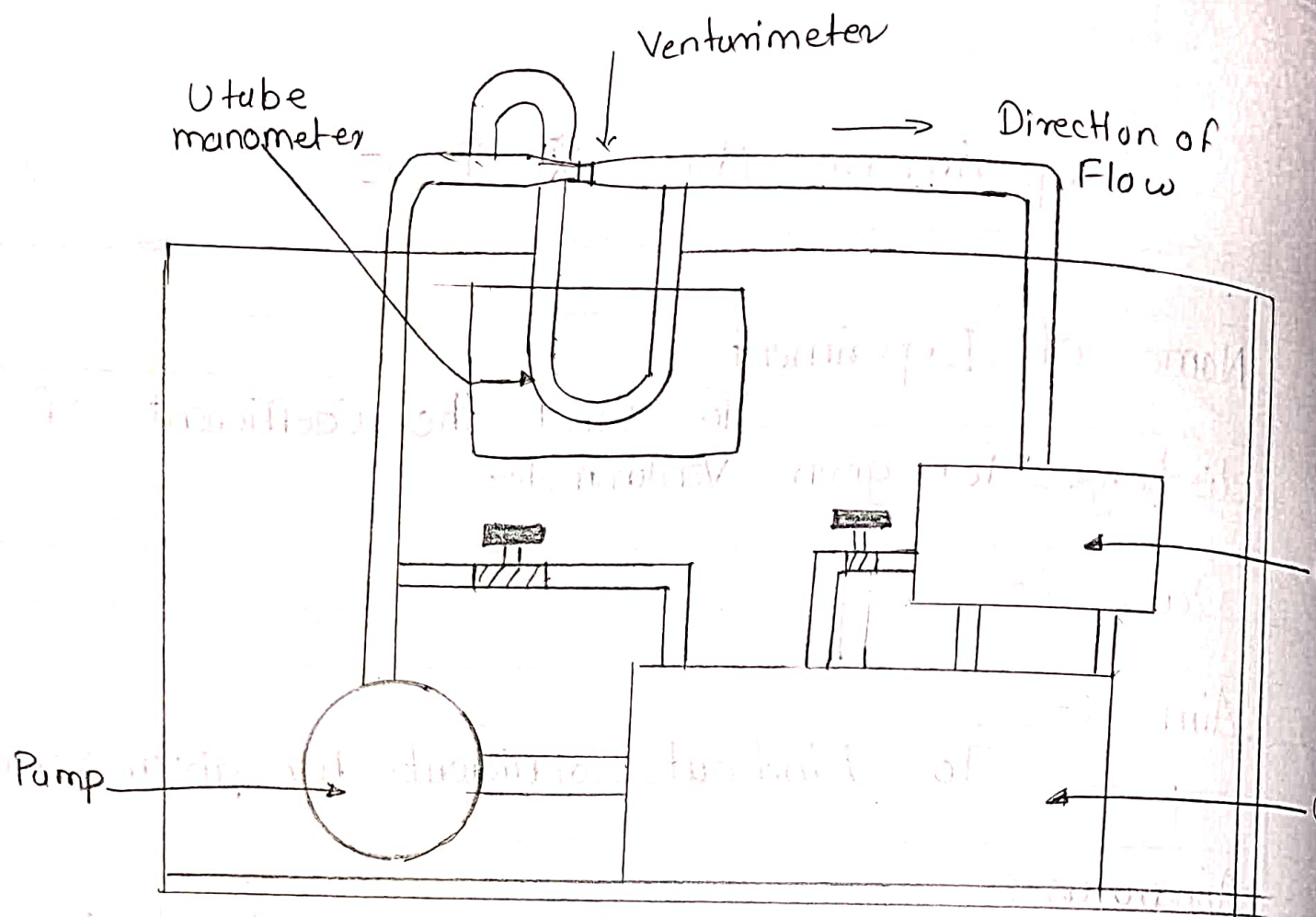
Venturimeter is a device used to measure the rate of flow through pipe. Its constructed by using Bernoulli's principle.

It Consist

- 1) Convergent inlet section
- 2) Throat section
- 3) Divergent section.

Discharge Calculated By

$$Q_{act} = \frac{C_d a_1 a_2}{\sqrt{a_1^2 - a_2^2}} \sqrt{2gH}$$



Observation :-

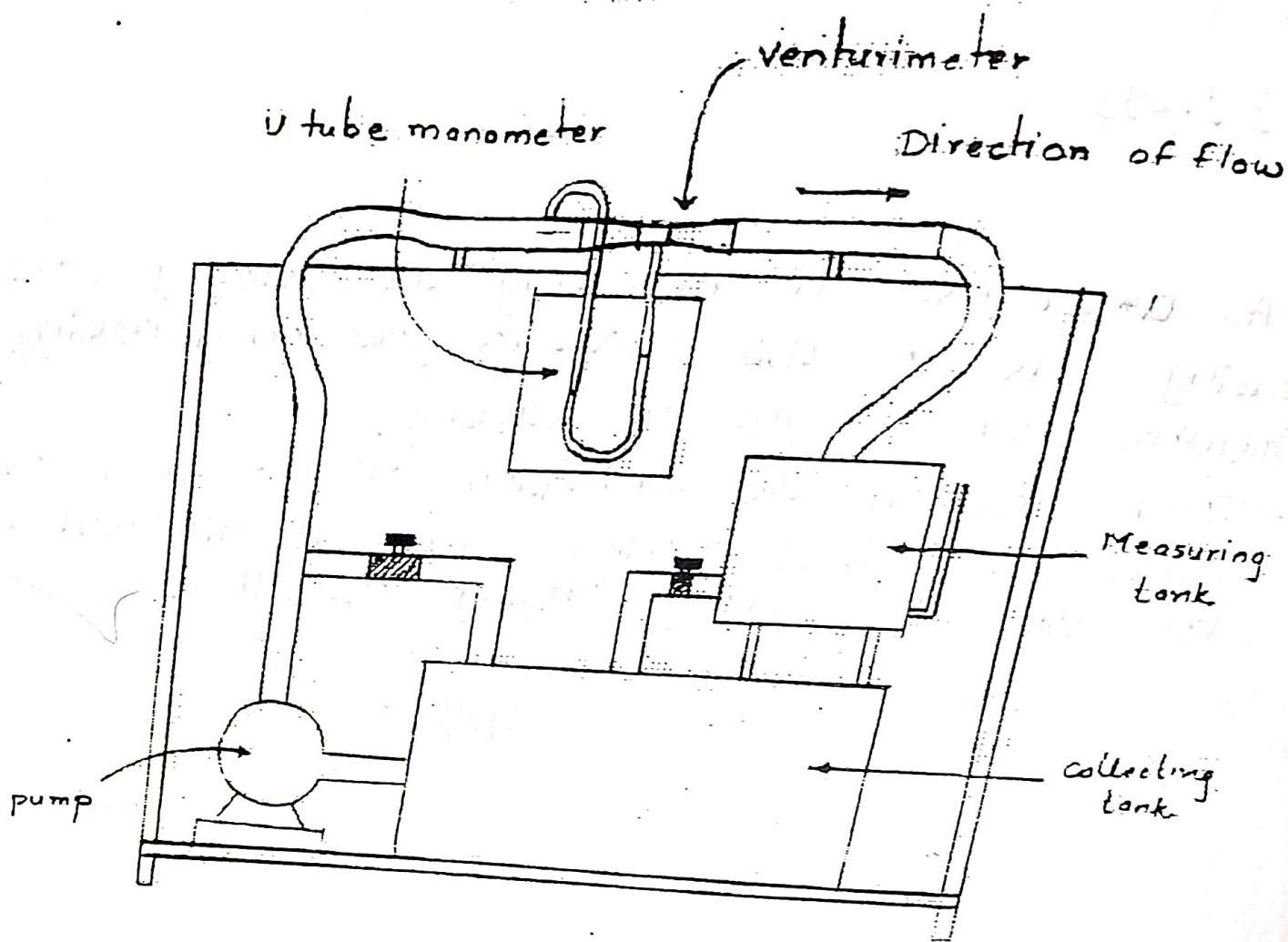
$$L = 41 \text{ cm} = 0.41 \text{ m} \quad B = 0.325 \text{ m}$$

$$\text{Measuring tank Size} = l \times b \times h = 0.0133 \text{ m}^3$$

$$H = 0.1 \text{ m} \quad d_1 = 25.5 \text{ mm} = 0.0255 \text{ m}$$

$$d_2 = 0.0135 \text{ m} \quad A_1 = \frac{\pi}{4} d_1^2 = 5.10 \times 10^{-4} \text{ m}^2$$

$$A_2 = \frac{\pi}{4} d_2^2 = 1.43 \times 10^{-4} \text{ m}^2$$



Where,

$a_1 =$ Area of inlet m^2

$a_2 =$ Area of outlet m^2

$H =$ Pressure head difference betⁿ inlet & outlet. throat.

$$= x \left(\frac{S_m - 1}{S_L} \right) \text{ if } S_m > S_L$$

Procedure.

1. Start the pump & by using valve maintain the Flow through pipe & venturimeter.
2. Measure inlet dia (d_1) & throat dia (d_2)
3. Measure measuring tank (l) & (h)
4. Remove air differential manometer pipes.
5. For each reading measure time (t) sec to Fill 10 cm of water in measuring tank & measure reading different (x) in m of both limbs of diff. manometer.

Different (x) in m of both limbs of diff. manometer.

6. Calculate Q_{act} . h. Q_{th} . cd . Q_s per given eqⁿ

7. Again adjust the flow for subsequent reading & follow the the step No 5.

Conclusion :-

From the experiment we have seen as well as study of venturimeter the avg. $cd = 0.93$ which is range 0.96 to 0.98.

Observation Table

Sr No	Time (sec)	Manometer reading (x) (cm)	Pressure difference $h = x \left(\frac{\rho_m}{\rho} - 1 \right)$	$Q_{th} = \frac{a_1 a_2 \sqrt{2gH}}{\sqrt{a_1^2 - a_2^2}}$ (m ³ /sec)	$Q_{act} = \frac{\text{Volume}}{\text{Time}}$ (m ³ /sec)	$C_d = \frac{Q_{act}}{Q_{th}}$
1	22.58	6.5	0.81	5.93×10^{-4}	5.90×10^{-4}	0.99
2	27	5	0.63	5.24×10^{-4}	4.93×10^{-4}	0.94
3	31.6	4.5	0.56	4.94×10^{-4}	4.21×10^{-4}	0.85

Calculations

$$Q_{act1} = \frac{\text{Volume}}{\text{Time}} = \frac{0.013325}{22.58} = 5.90 \times 10^{-4} \text{ m}^3/\text{sec}$$

$$Q_{act2} = \frac{\text{Volume}}{\text{Time}} = \frac{0.013325}{27} = 4.93 \times 10^{-4} \text{ m}^3/\text{sec}$$

$$Q_{act3} = \frac{\text{Volume}}{\text{Time}} = \frac{0.013325}{31.6} = 4.21 \times 10^{-4} \text{ m}^3/\text{sec}$$

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